

Multi-Scale Dynamical Approaches for Analyzing Brain Signals

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MDIG2026



THE UNIVERSITY OF
SYDNEY

What are we doing here today

1

Where I'm coming from: multiscale dynamics in the brain, and how we've traditionally analyzed them

2

Introducing the *pyspi* library for pairwise interactions: Theory and implementation

3

Interactive demo + Q&A for *pyspi*

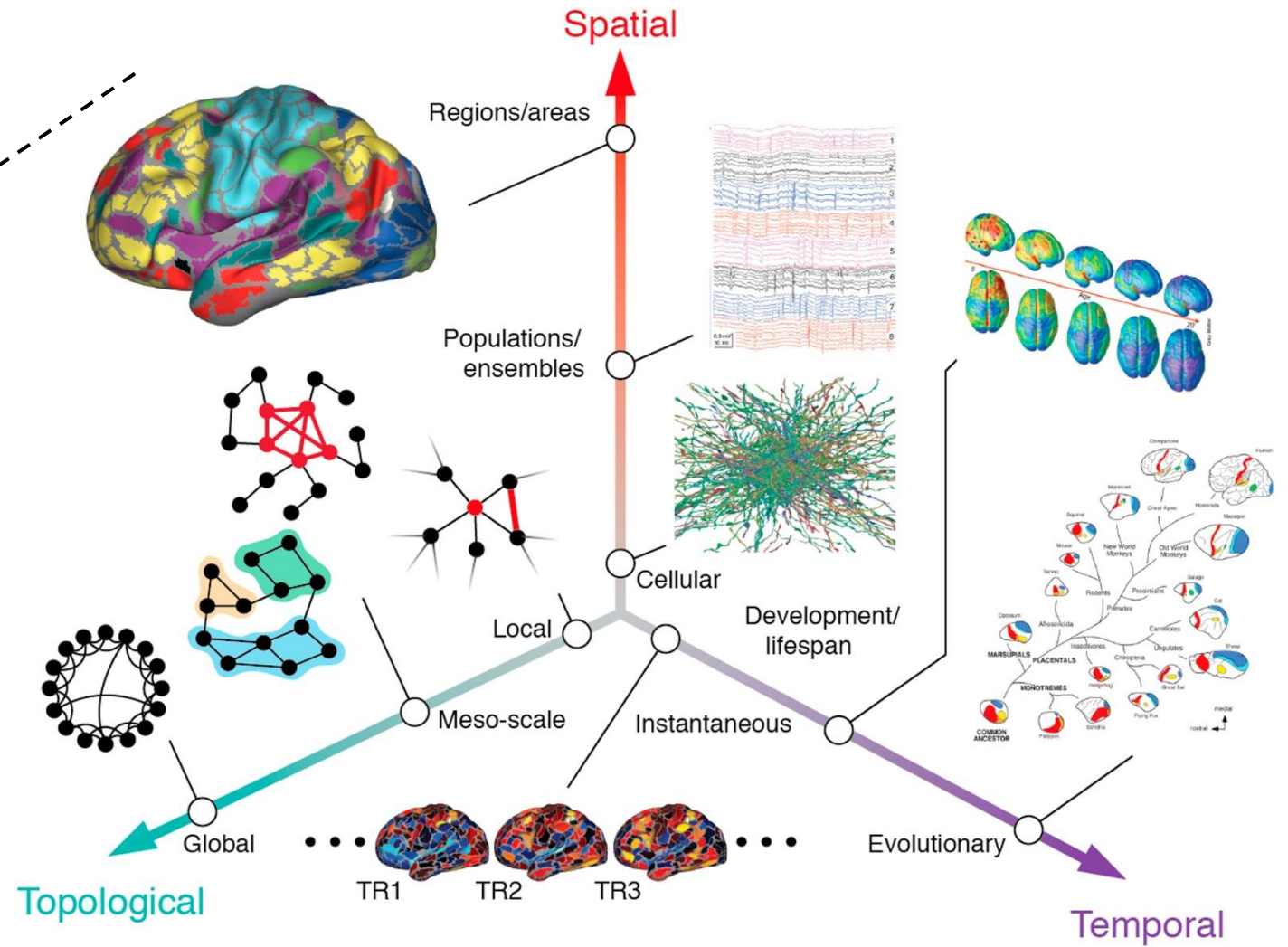
4

Take-home advice for your own analyses and examples of *pyspi* applications to brain dynamics

Multi-scale activity and interactions in the brain



Multi-scale activity and interactions in the brain



Betzel & Basset, *NeuroImage* (2016)

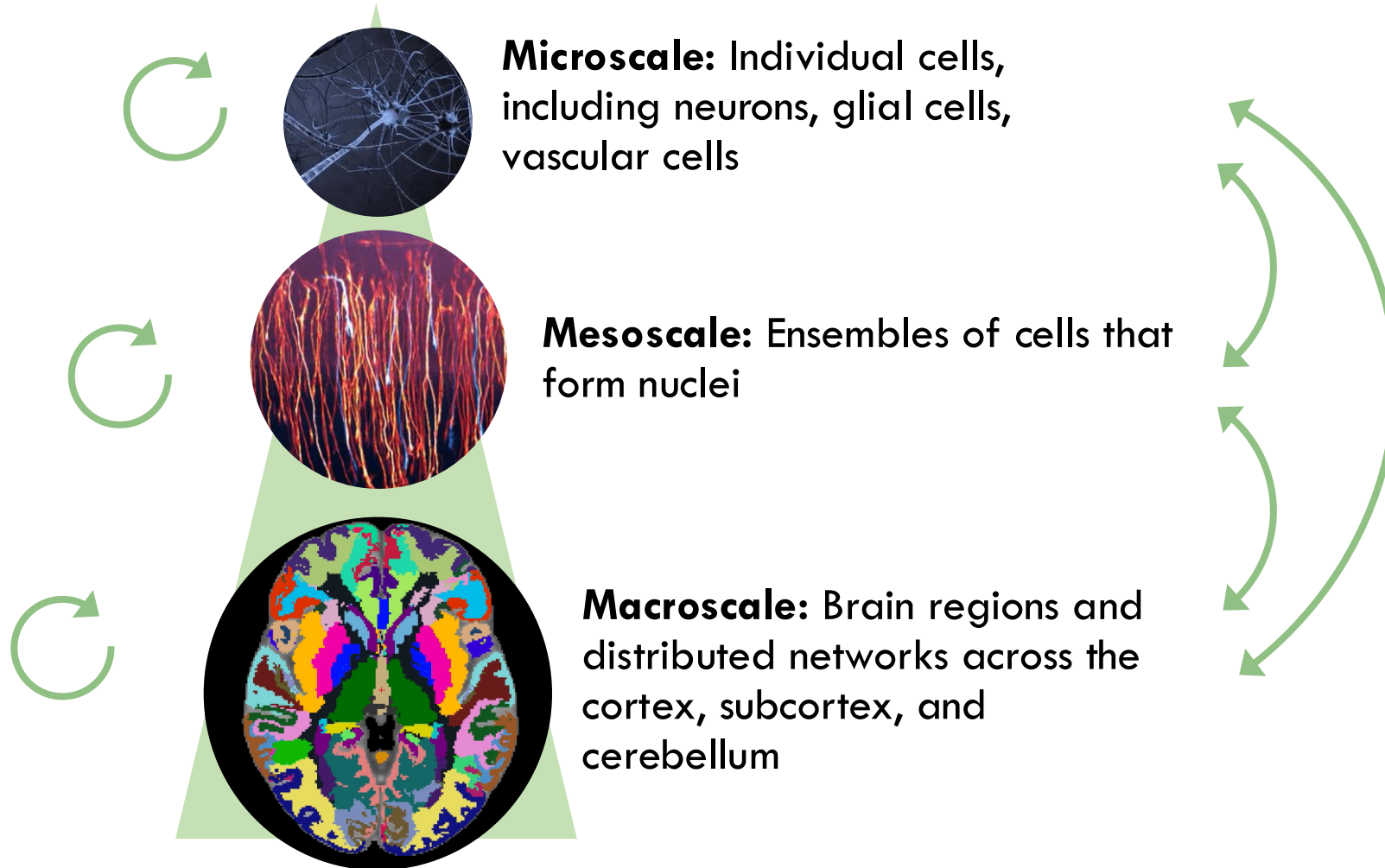
Multi-scale activity and interactions in the brain

Spatial scales in the brain

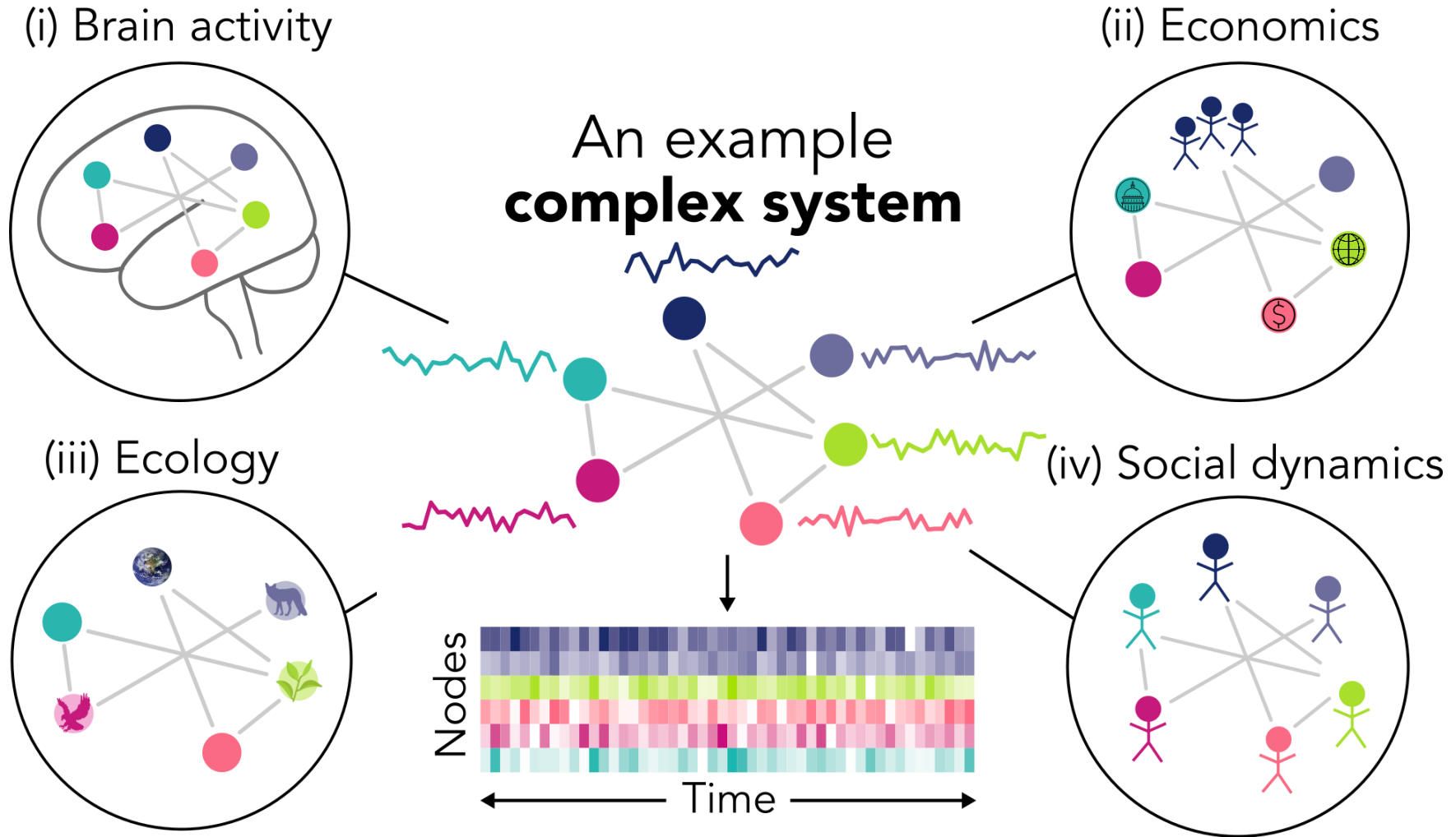
Harvard University

Dr. Alexandre Dumoulin

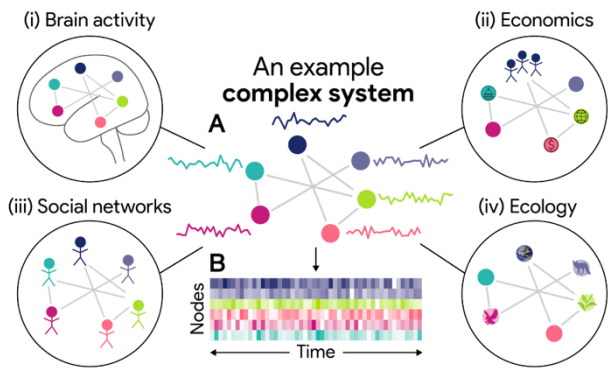
Weill Cornell



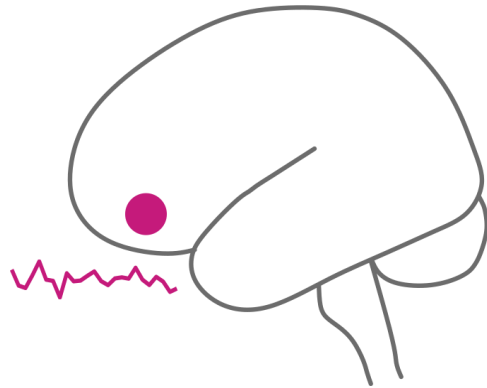
The brain as a **complex system**



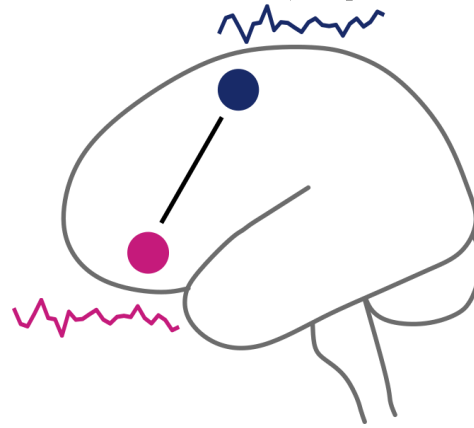
Multiple ways of parsing and quantifying brain activity



Local

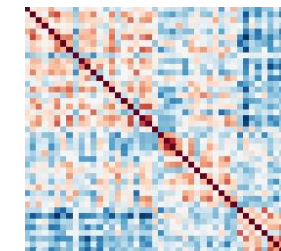
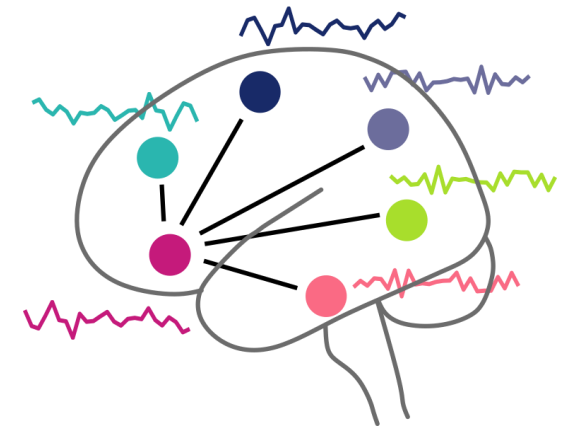


Pairwise (dyads)



all
pairs

Distributed



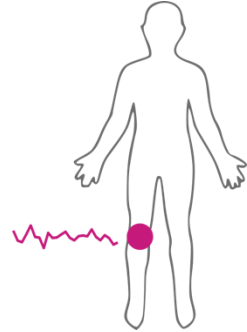
network
properties

Many applications across neuroscience + **social dynamics**

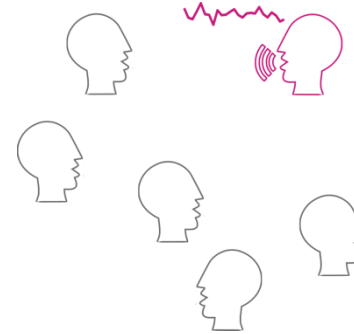
Brain activity



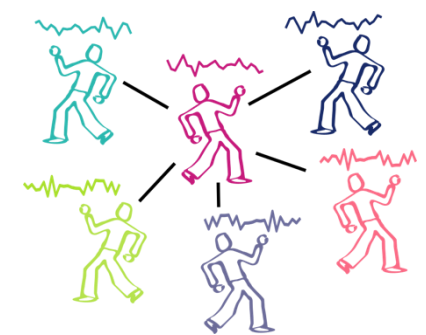
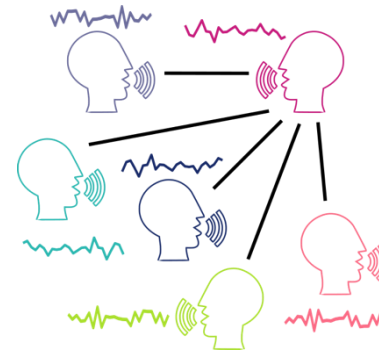
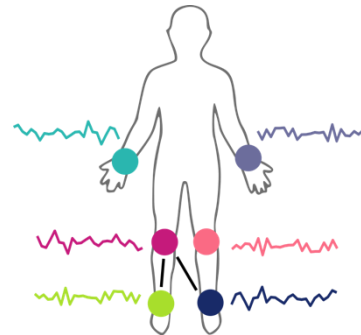
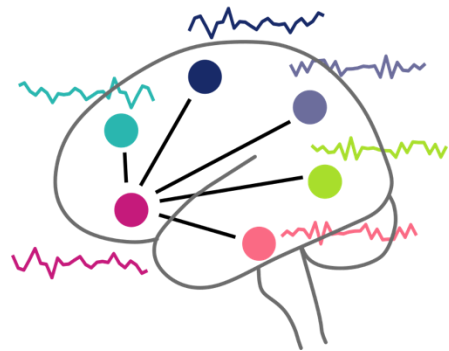
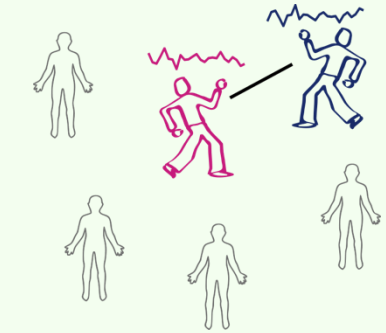
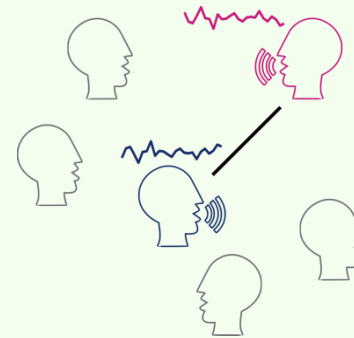
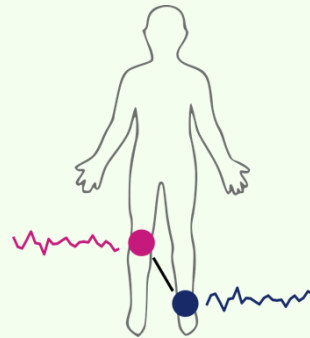
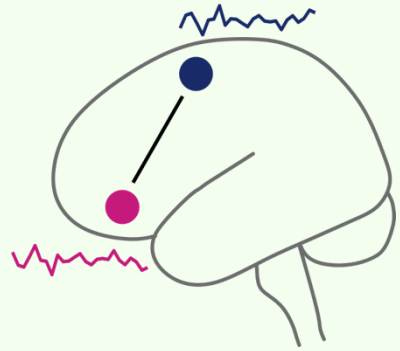
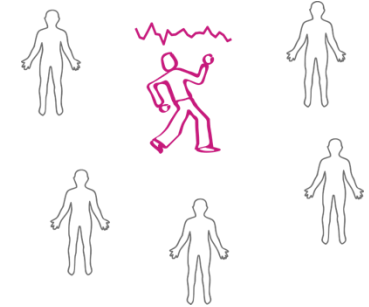
Individual kinematics



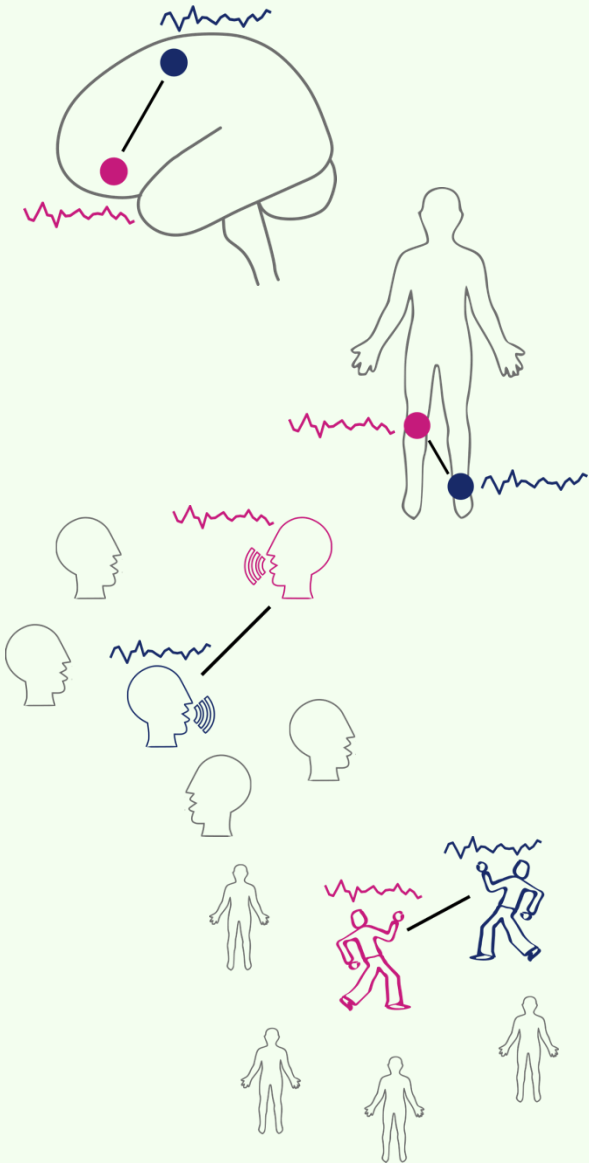
Conversation dynamics



Social movement: dancing



Today's focus: Quantifying pairwise (dyad) interactions

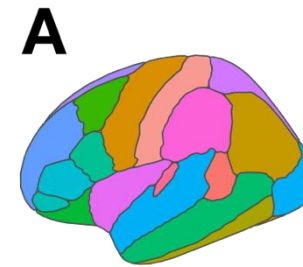
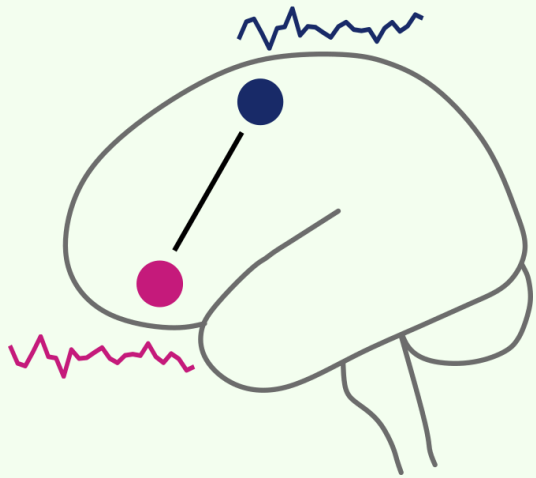


How does information flow between the two time series?

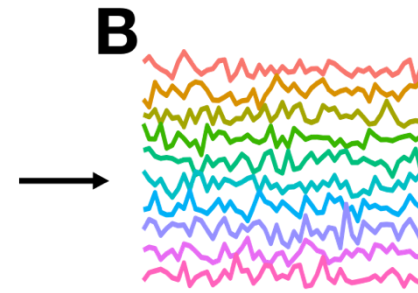
Do the two systems exhibit statistical dependencies in the time and/or frequency domains?

Does the signal in one time series predict that of the other time series, using linear and/or nonlinear modelling?

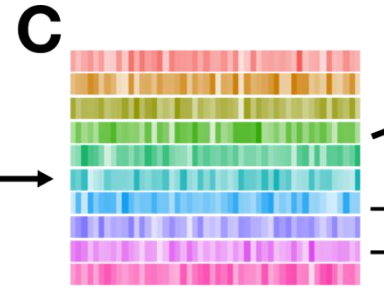
Conventional neuroscience application: Pearson correlation



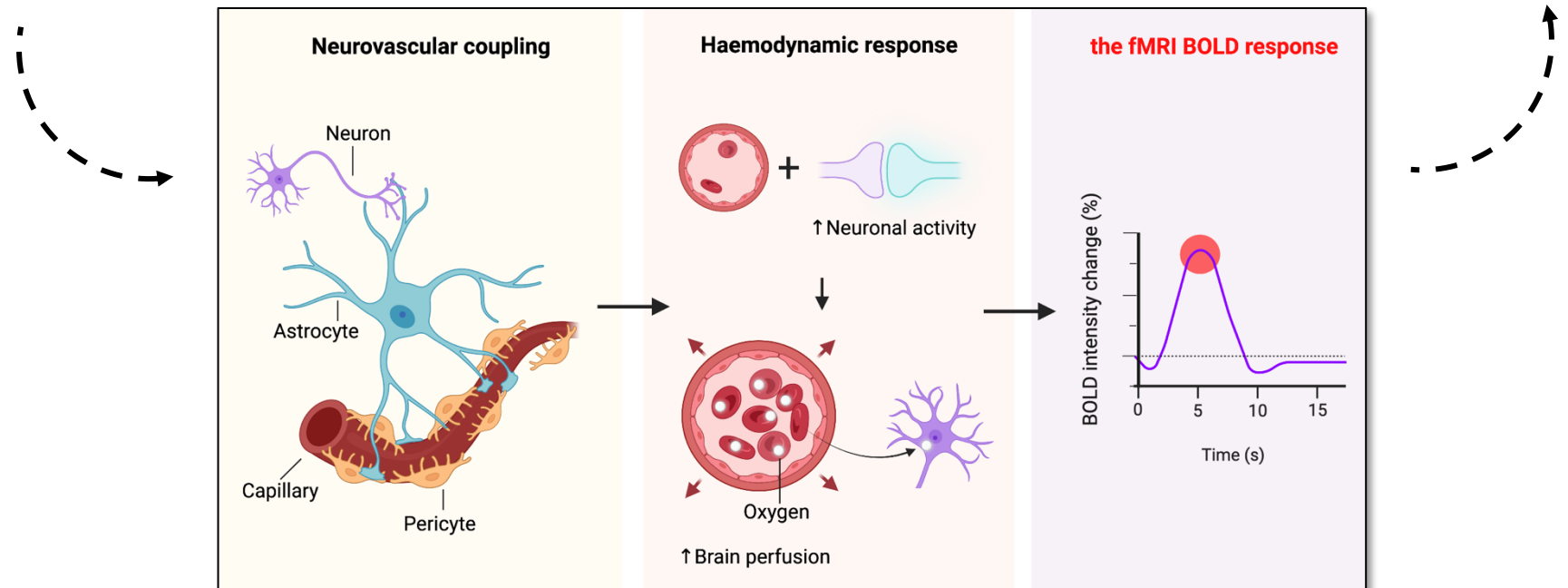
Functional volume



Extract activity

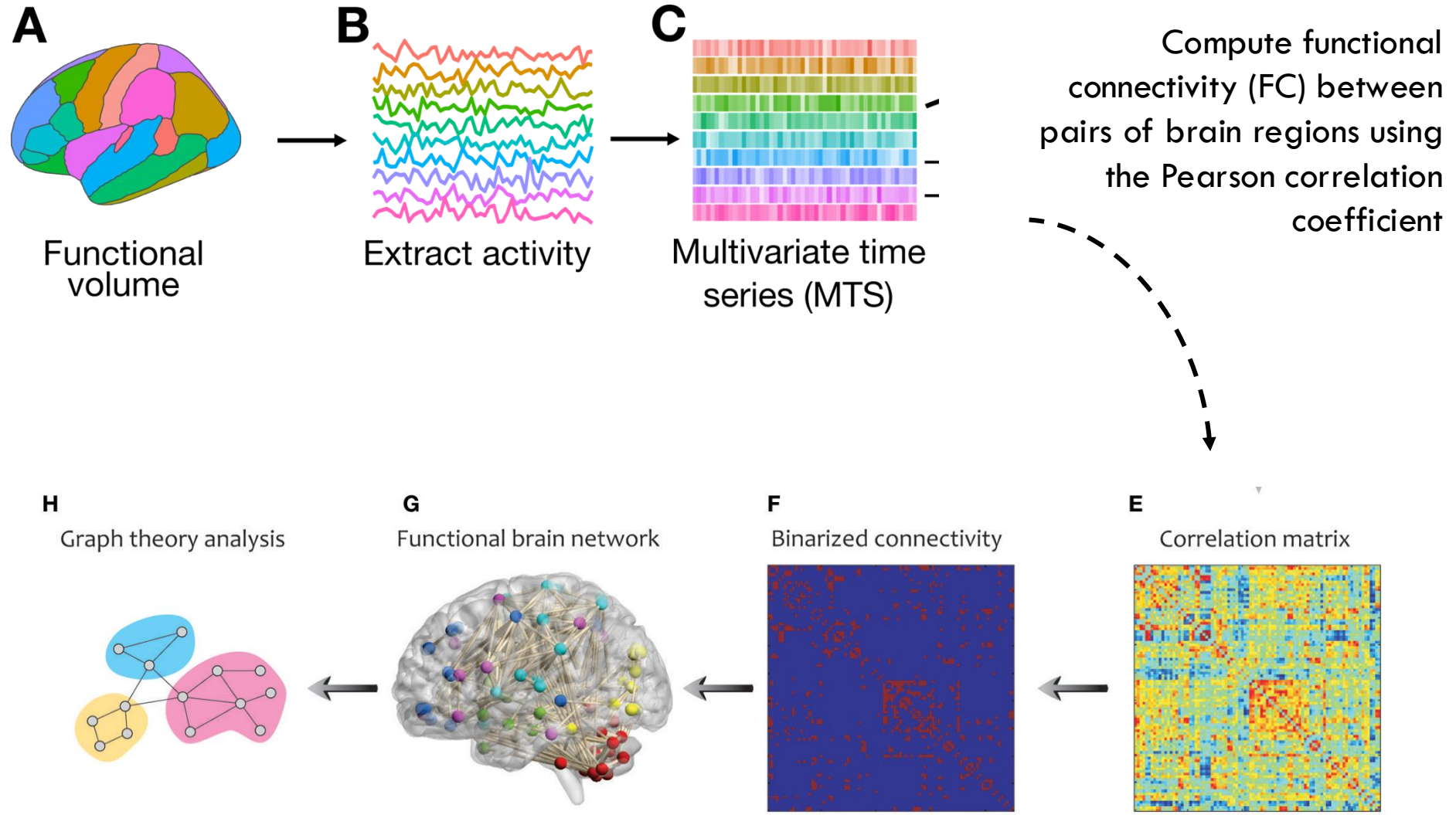
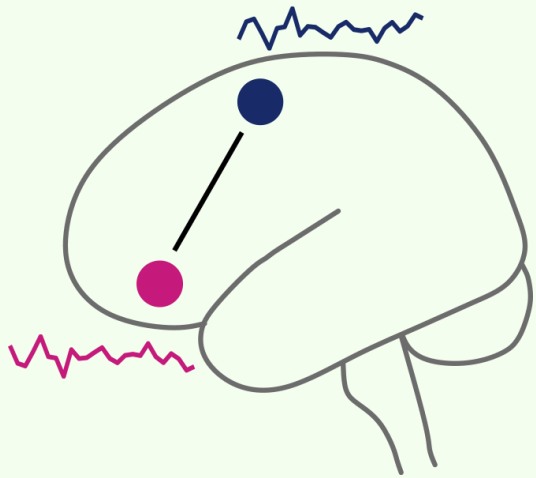


Multivariate time series (MTS)



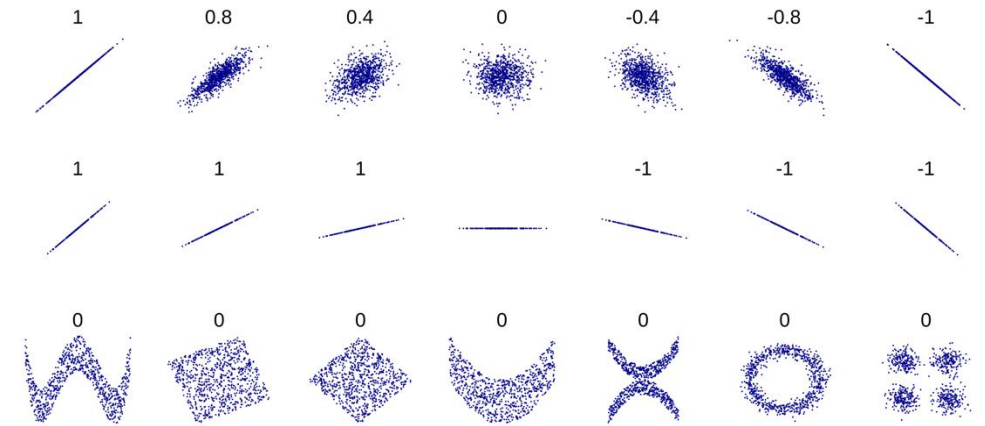
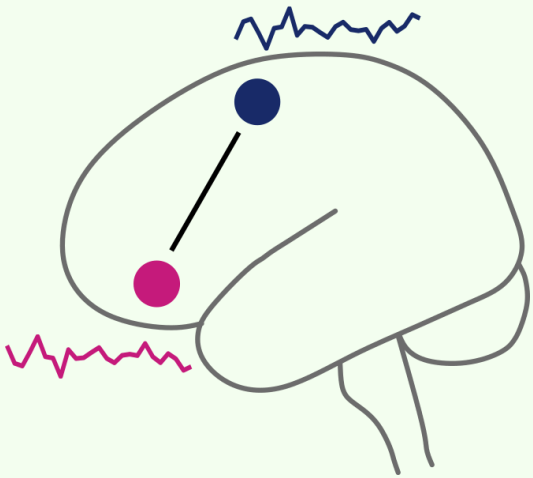
Source: bioRender (Tamara Vasilkovska)

Conventional neuroscience application: Pearson correlation



Source: Farahani et al. *Frontiers in Neuroscience* (2019)

Potential pitfalls of the Pearson correlation coefficient



Source: Wikipedia ('Pearson correlation coefficient')

Assumes a joint bivariate Gaussian distribution for valid statistical inference

Does not always hold for empirical brain-imaging data

Sensitive to linear relationships

Cannot detect nonlinear relationships (monotonic or non-monotonic)

Focuses on instantaneous, time-locked associations

Misses time-lagged associations

Sensitive to outliers

Not robust to a small number of extreme values

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The origins of pyspi

Article | Published: 25 September 2023

Unifying pairwise interactions in complex dynamics

[Oliver M. Cliff](#), [Annie G. Bryant](#), [Joseph T. Lizier](#), [Naotsugu Tsuchiya](#) & [Ben D. Fulcher](#) 

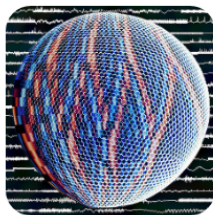
[Nature Computational Science](#) **3**, 883–893 (2023) | [Cite this article](#)

 [Save article](#)

5224 Accesses | **77** Citations | **138** Altmetric | [Metrics](#)

DynamicsAndNeuralSystems/**pyspi**

Comparative analysis of pairwise interactions in multivariate time series.



6

Contributors



10

Used by



252

Stars



34

Forks



Dr. Oliver M. Cliff



Dr. Annie Bryant



A/Prof Joe Lizier



Prof Nao Tsuchiya

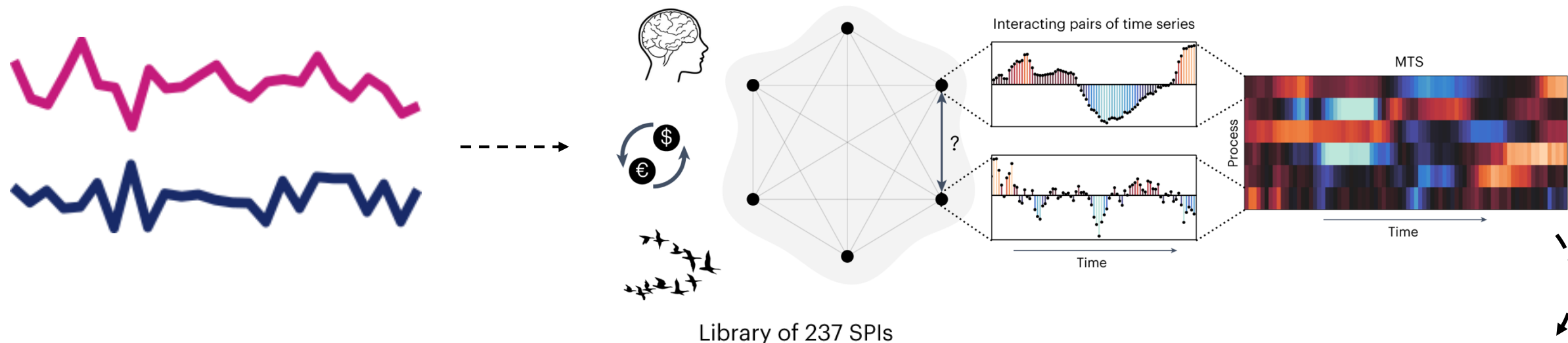


A/Prof Ben Fulcher



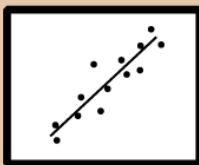
Joshua Moore

A unified library of 237* statistics for pairwise interactions



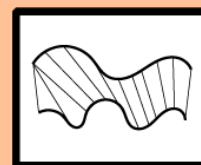
Basic (21 SPIs)

Covariance
Kendall's τ
Cross-correlation
...



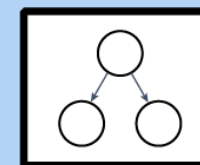
Distance similarity (26 SPIs)

Distance correlation
Heller-Heller-Gorfine test
Dynamic time warping
...



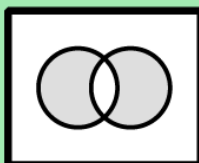
Causal indices (10 SPIs)

Additive noise models
Convergent cross-mapping
...



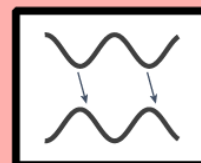
Information theory (37 SPIs)

Mutual information
Transfer entropy
Integrated information
...



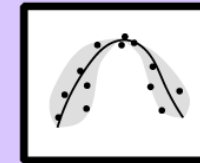
Spectral (126 SPIs)

Coherence magnitude
Directed coherence
Spectral Granger causality
...



Miscellaneous (17 SPIs)

Linear model fits
Cointegration
Envelope correlation
...



A unified library of 237* statistics for pairwise interactions

M1: mean phase lag/slope indices and group delay

M2: causal models

M3: directed information and causal entropy measures

M4: transfer entropy

M5: parametric granger causality and integrated information

M6: parametric granger causality and directed spectral measures

M7: max phase lag indices

M8: phase slope indices (wavelet)

M9: mean of barycenters

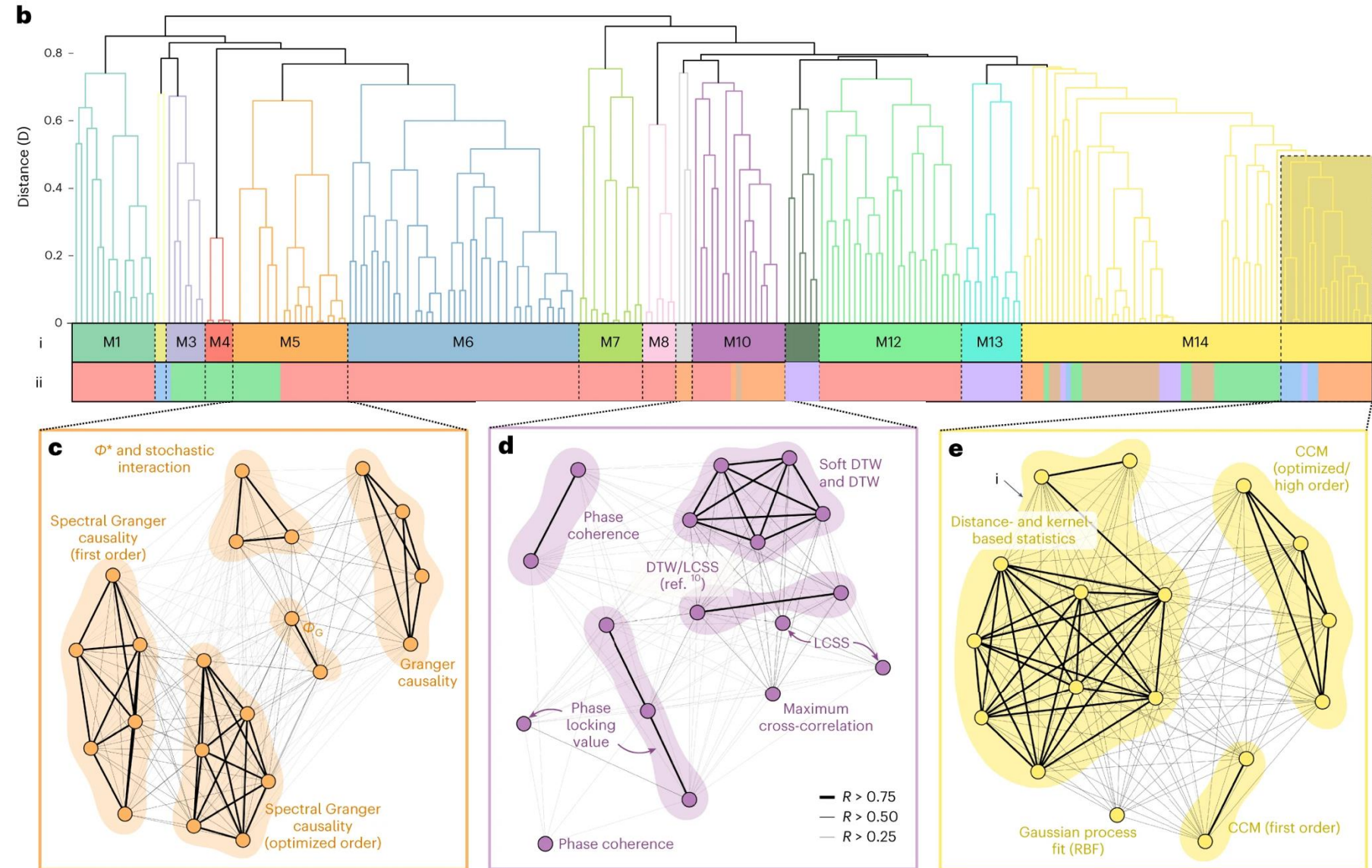
M10: dynamic time warping, phase coherence and locking values

M11: Power envelope correlation

M12: undirected spectral measures

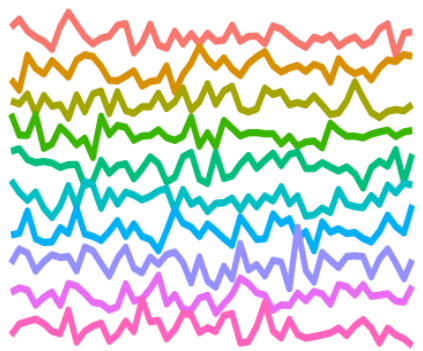
M13: co-integration

M14: a mix of contemporaneous linear-dependence statistics, information-theoretic measures, convergent cross-mapping, maximum barycenters, distance- and kernel-based statistics



Cliff et al. *Nat Comp Sci* (2023)

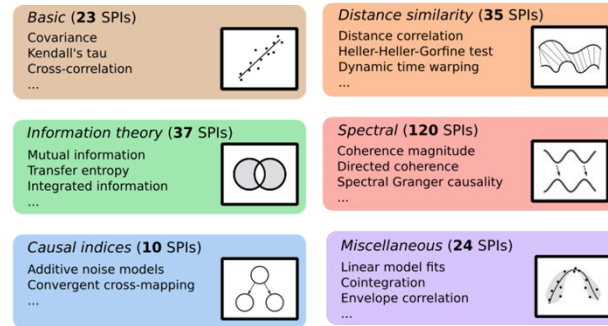
Standard pipeline for pyspi-based analysis



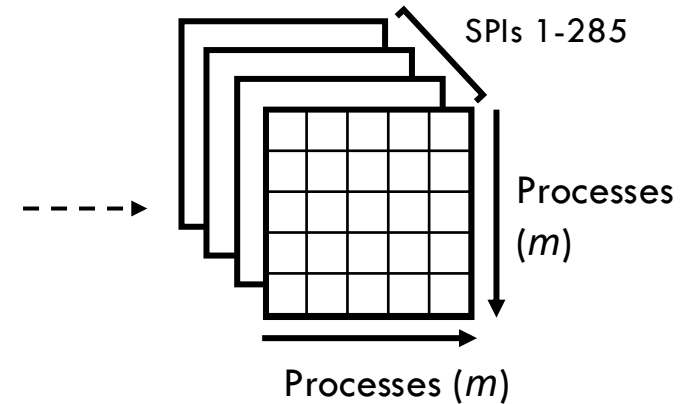
Input m -dimensional
time series
(numpy.array)



(Optionally) detrend and/or
normalize time series; load
into a calculator object

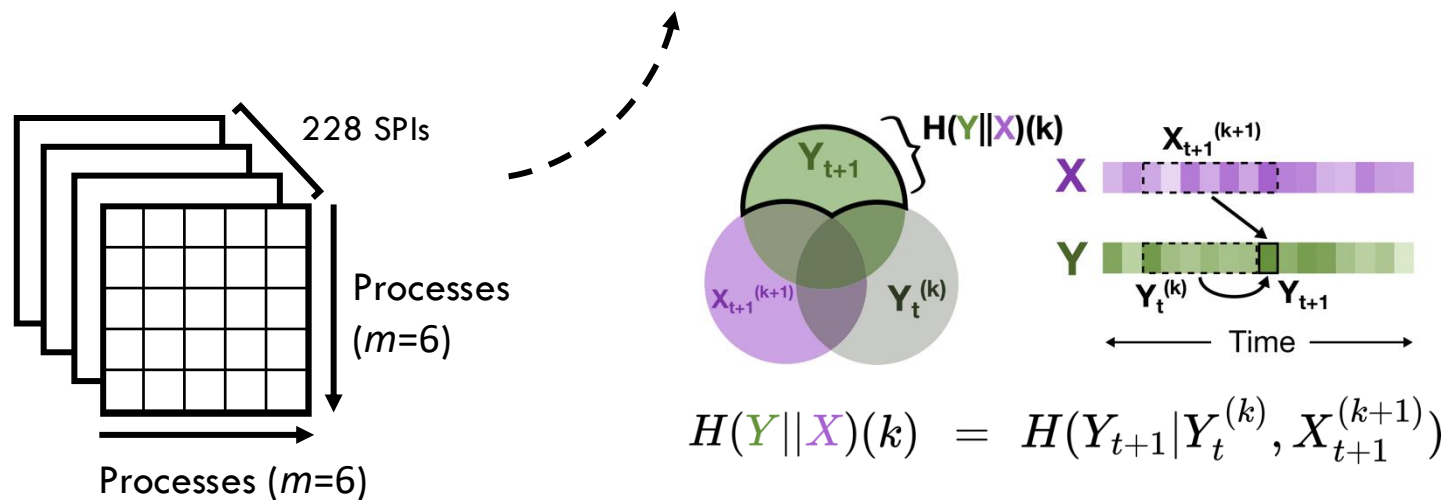
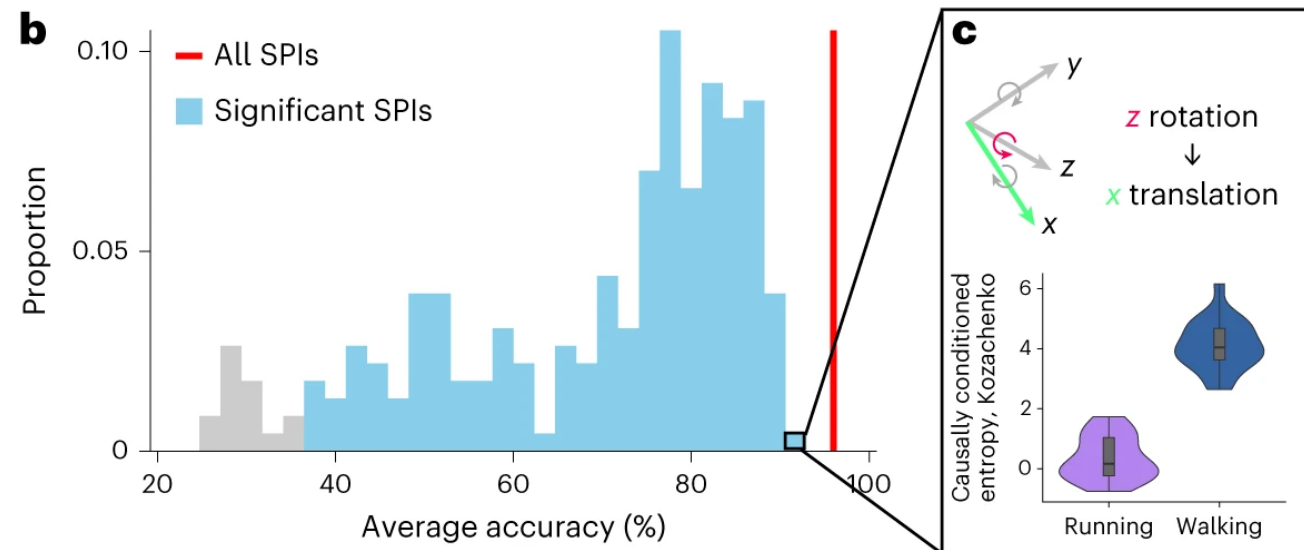
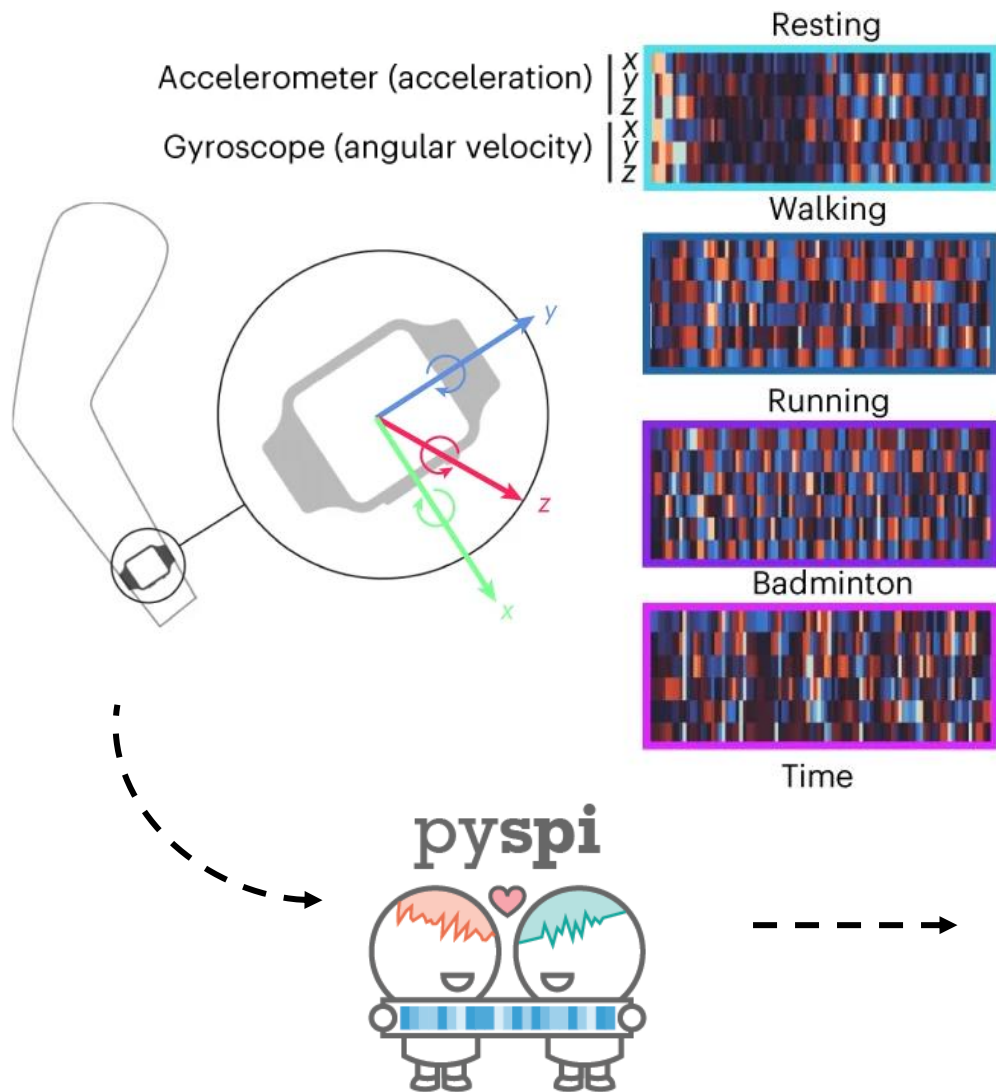


Compute up to 285 **statistics**
for pairwise interactions
(SPIs) with
`calculator.compute()`



Output: a set of tabular
dataframes containing
pairwise values for each pair
of processes by SPI
(`calculator.table`)

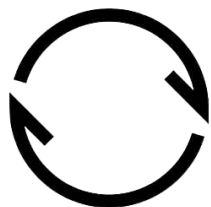
The OG pyspi case study: Smart watch actigraphy



Cliff et al. *Nat Comp Sci* (2023)

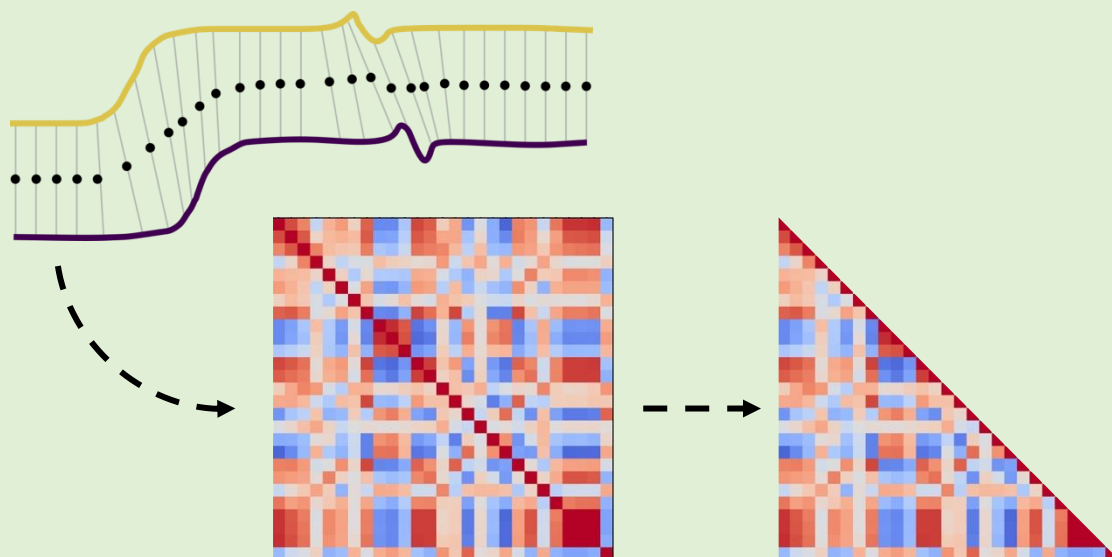
Bryant et al. *J Royal Soc Int* (2026)

Practical considerations for analysis with pyspi



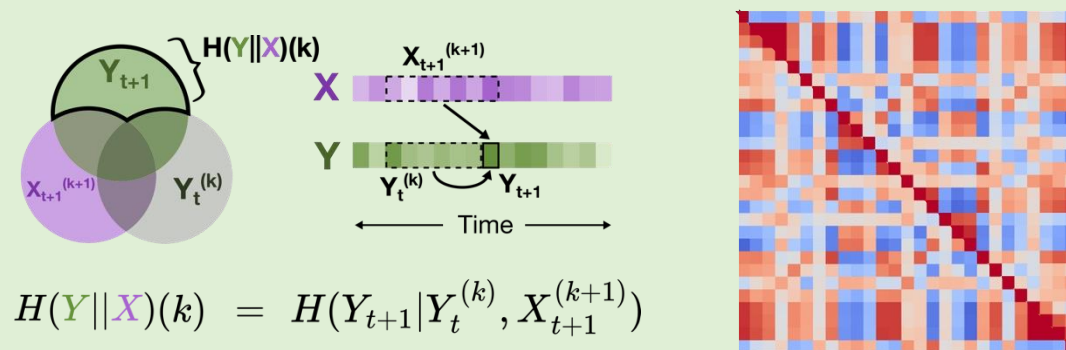
Some SPIs are **undirected**, meaning they capture a perfectly symmetric relationship in both directions. However, some SPIs are **directed**, meaning values can differ from $A \rightarrow B$ versus $B \rightarrow A$.

Undirected example: Dynamic time warping



Only need to keep lower half of SPI matrix

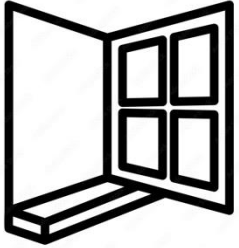
Directed example: Causally conditioned entropy



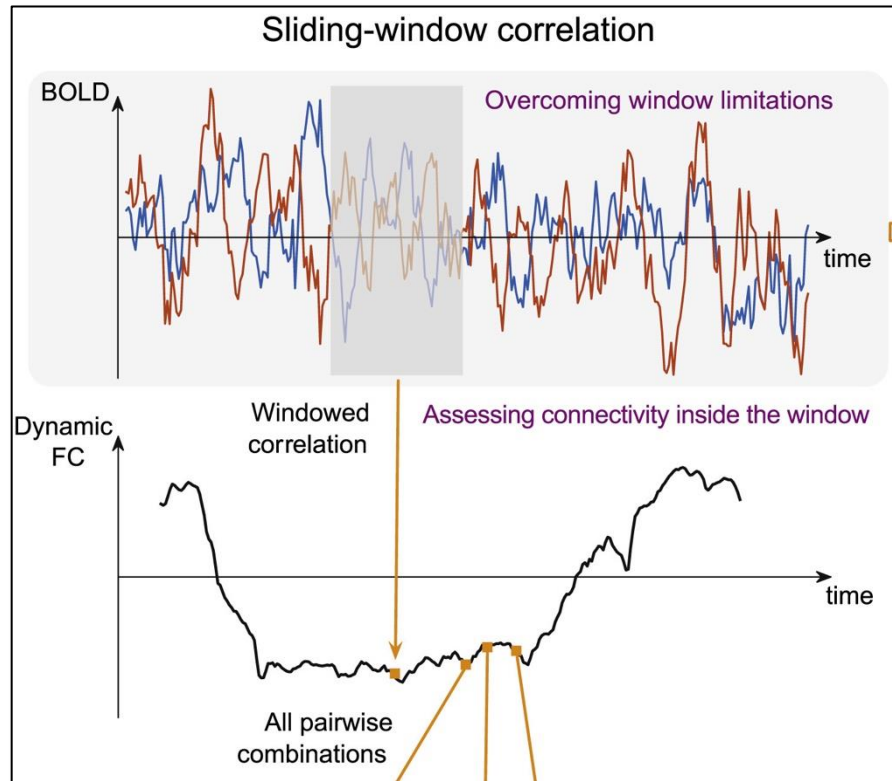
$$H(Y||X)(k) = H(Y_{t+1}|Y_t^{(k)}, X_{t+1}^{(k+1)})$$

Upper and lower triangles are different, so we need to keep the full SPI matrix

Practical considerations for analysis with *pyspi*



While some SPIs inherently apply windowing as part of their computation, each SPI yields one **static** measure representing the entire duration of the bivariate time series; in other words, **SPIs are not time-varying**.



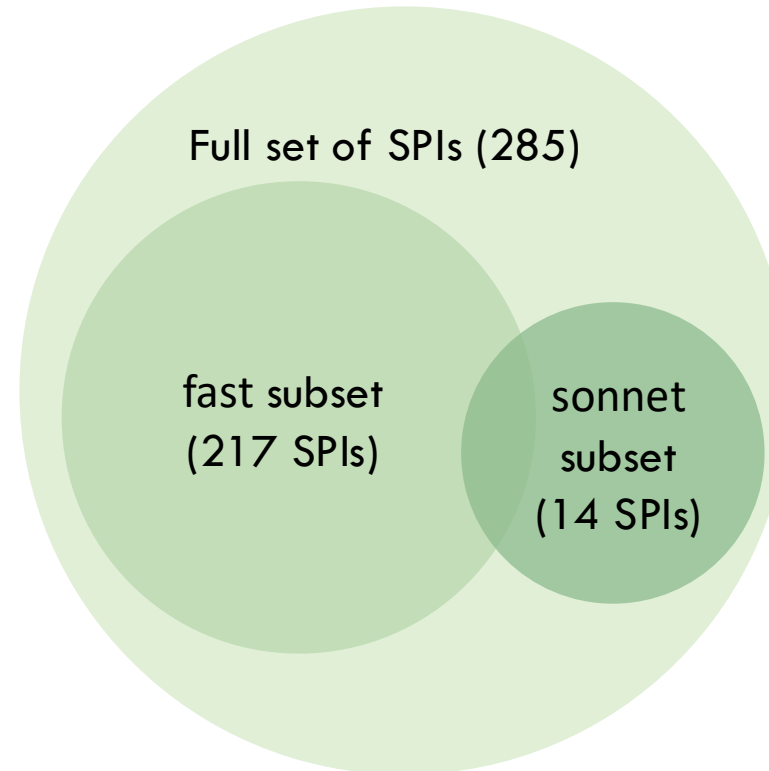
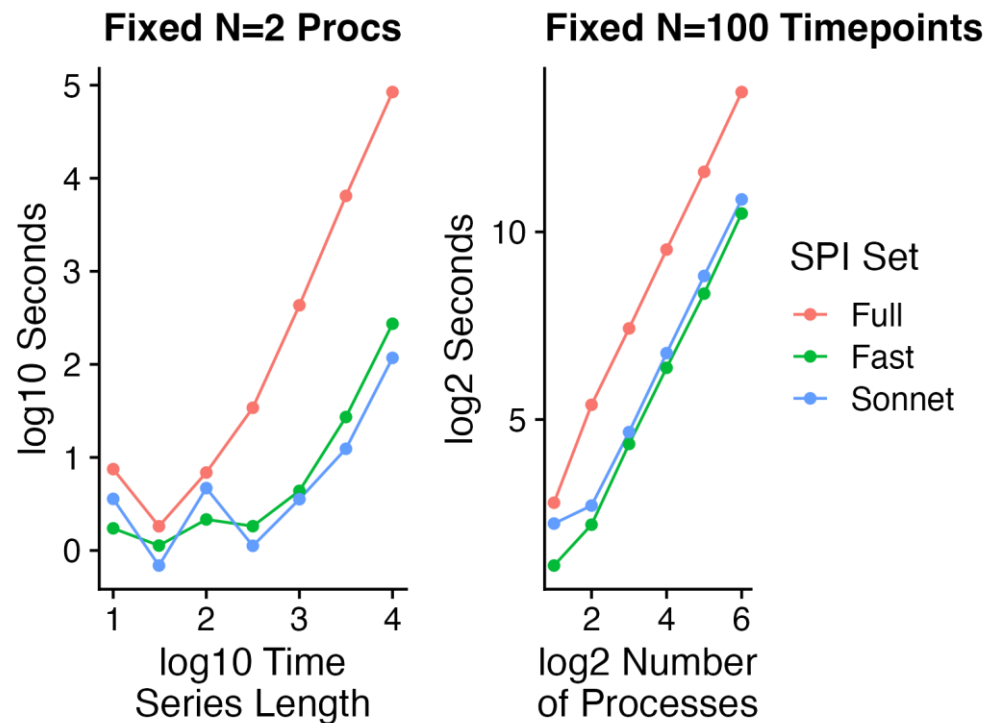
Preti et al. *NeuroImage* (2017)

If you have a hypothesis about a **time-varying relationship** between elements in a complex system, currently, you will need to implement your time-varying method of choice (windowing, state modelling, etc.) and compute each desired SPI per temporal subset.

Practical considerations for analysis with pyspi

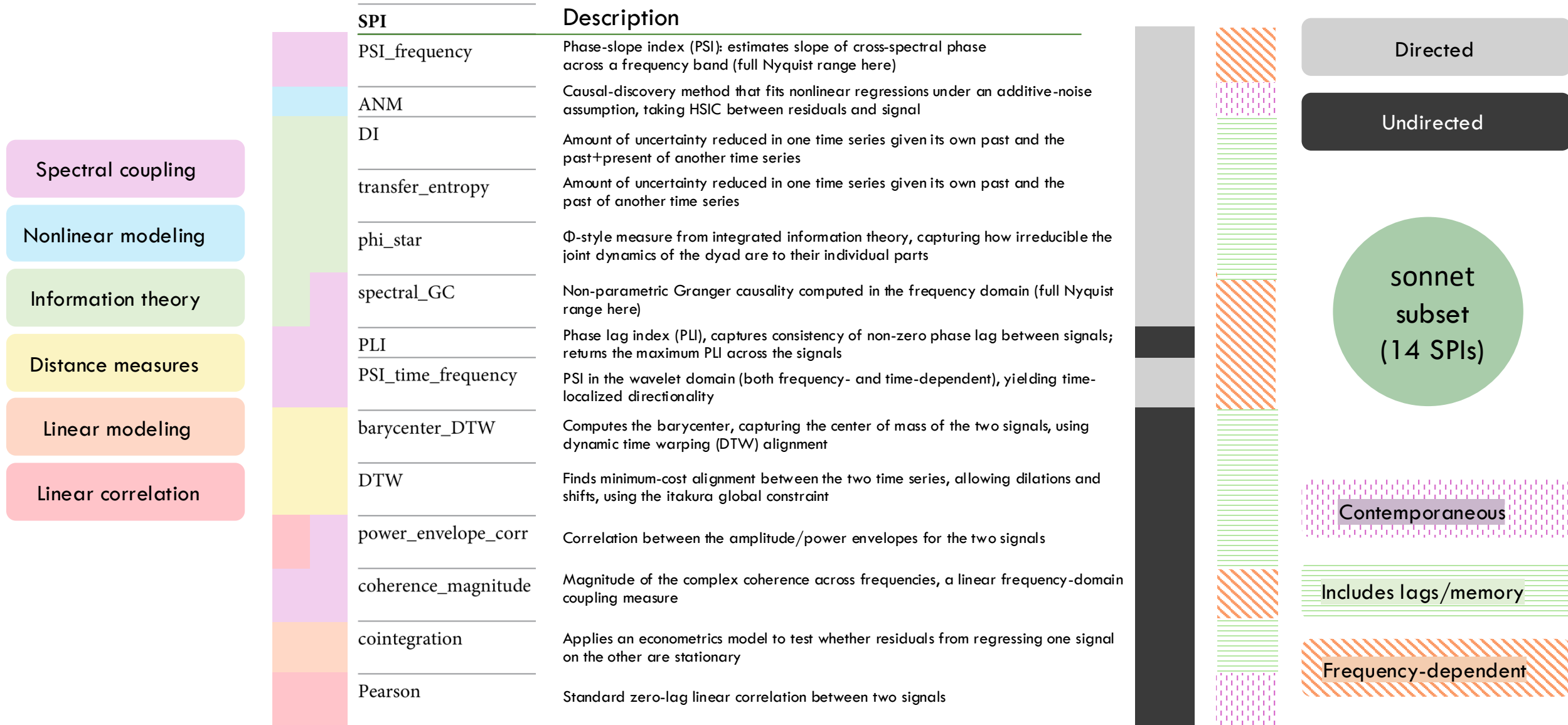


The **time** and **computational resources** can both increase rapidly with increasing number of processes, m . Additionally, some SPIs take considerably longer than others and scale differently according to time series length and m .



In our tutorial, we will work with the fast and sonnet subsets for computational tractability.

Composition of the sonnet SPI subset



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Key links + details

General pyspi info



DynamicsAndNeuralSystems/pyspi



Python 3.9+

Tutorial: To run locally on your computer



https://github.com/anniegbryant/MDIG2026_pyspi_tutorial/blob/main/MDIG2026_pyspi_tutorial.ipynb



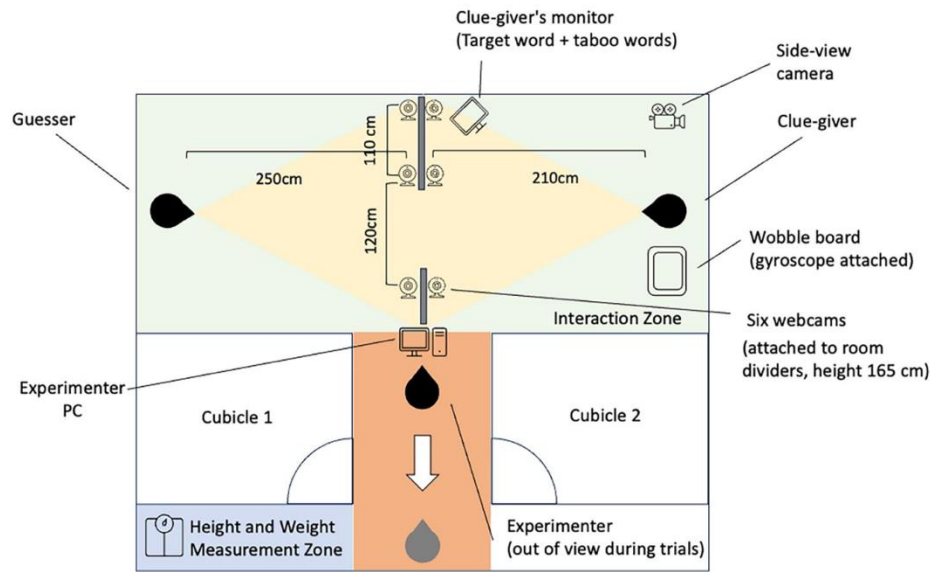
Tutorial: To run online using Google Colab



https://colab.research.google.com/drive/1TjJBY__V-DblAbeonb0LnocFIFT8l8_L?usp=sharing



MDIG2026 workshop application: BalanceCorpus data



BalanceCorpus dataset from
Li et al. *JASA* (2026)

Dyad: 103 + 203

Trial number: 12

Target word: Balloon 📌

Clue-giver condition: Balance board

Data type: Body

Camera: cam_01

We'll work up through five layers of
incremental complexity using pyspi:

1

One bivariate time series and one statistic for pairwise interactions (SPI)

2

Same bivariate time series as (1), using the sonnet subset of 14 SPIs

3

Same bivariate time series as (1), using the fast subset of 217 SPIs

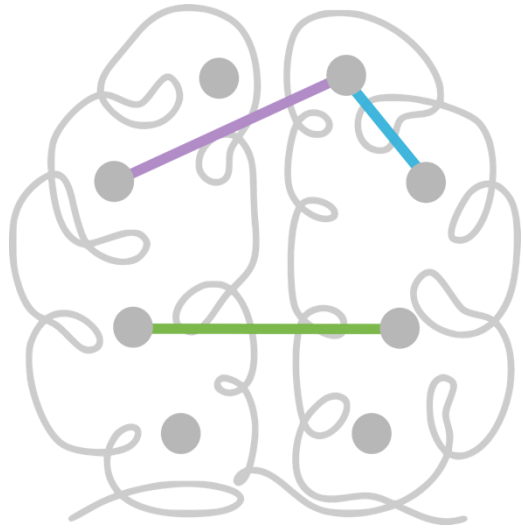
4

Four individual sets of bivariate time series, using the fast subset of 217 SPIs

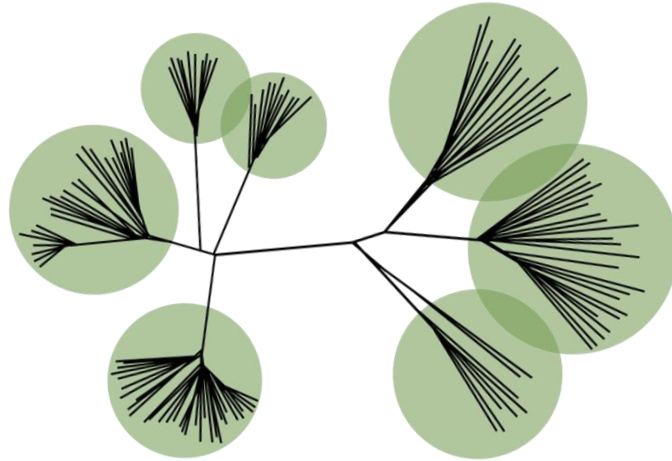
5

A 4-variable time series, using the sonnet subset of 14 SPIs

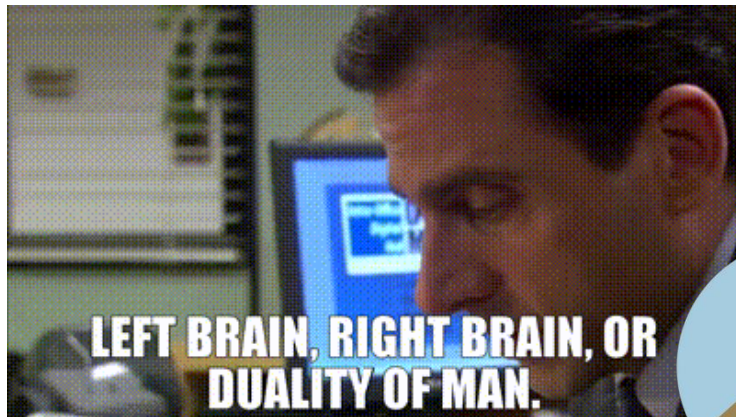
Brain-imaging application: Homotopic connectivity



Homotopic connectivity



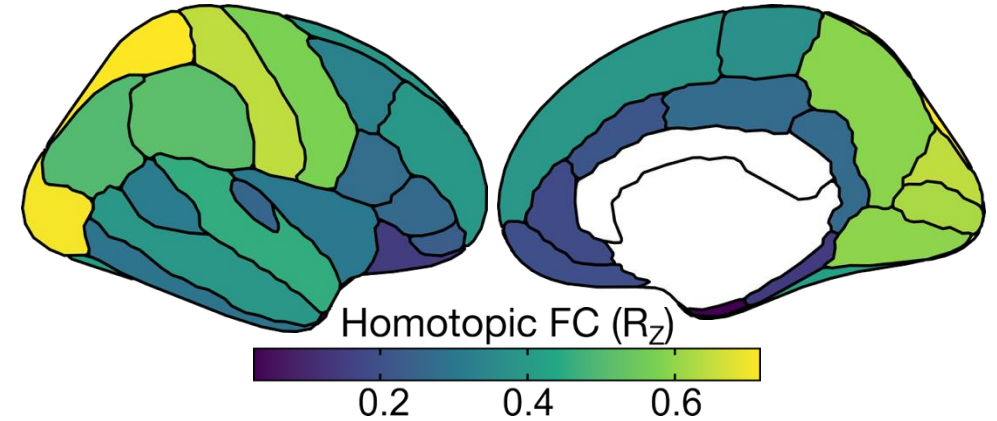
Observed across
numerous species:



Macaque

Rodent

Zebra finch



We'll apply pyspi to compute the sonnet set of SPIs across 34 pairs of homotopic brain regions (in one example participant from the Human Connectome Project)



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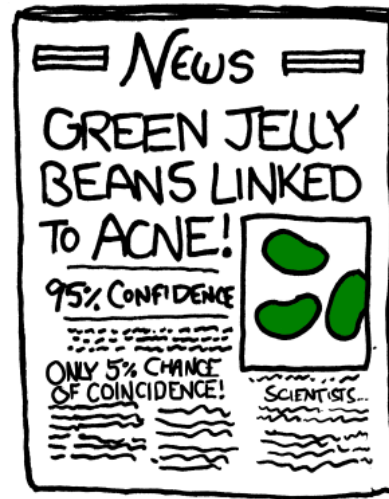
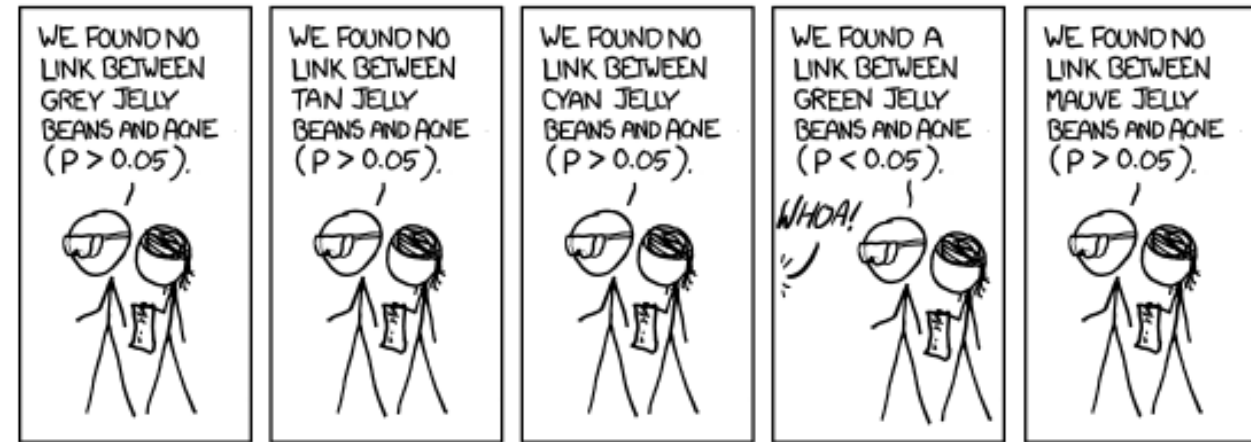
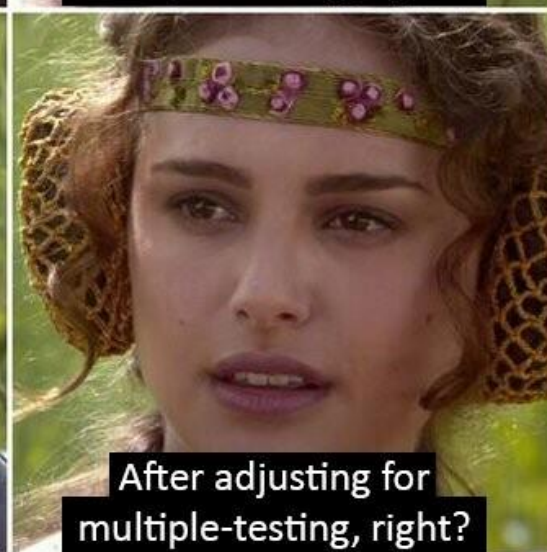
3

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Statistical considerations: Multiple comparisons correction



<https://xkcd.com/882>

MULTIPLE TESTING TOOLBOX

dealing with the MULTIPLE COMPARISONS PROBLEM

FWER

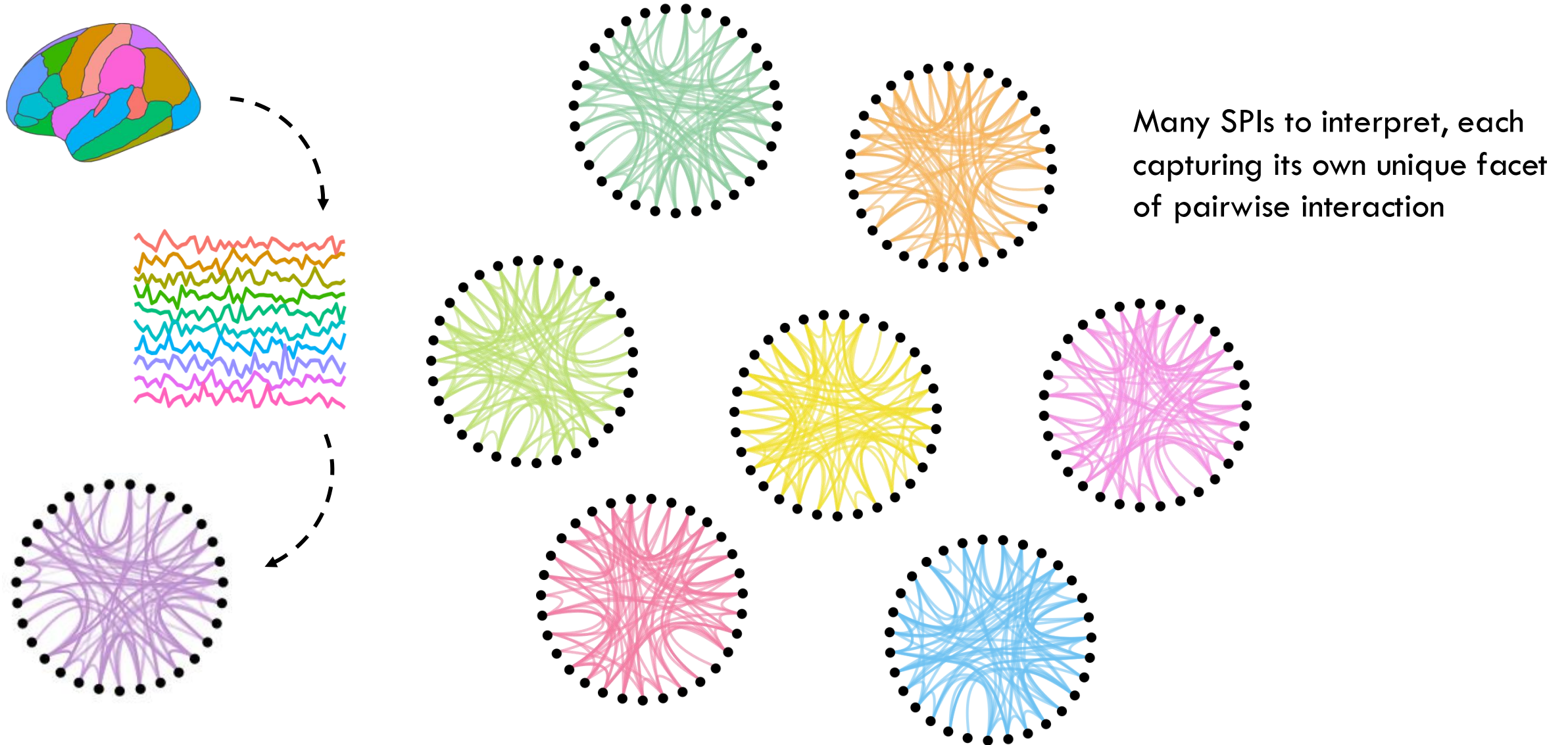
Bonferroni
Šidák
Holm-Bonferroni

FDR

Benjamini-Hochberg
Benjamini-Yekutieli
Storey's pFDR
Combined Prob. Fisher

Víctor Martínez-Cagigal (MATLAB)

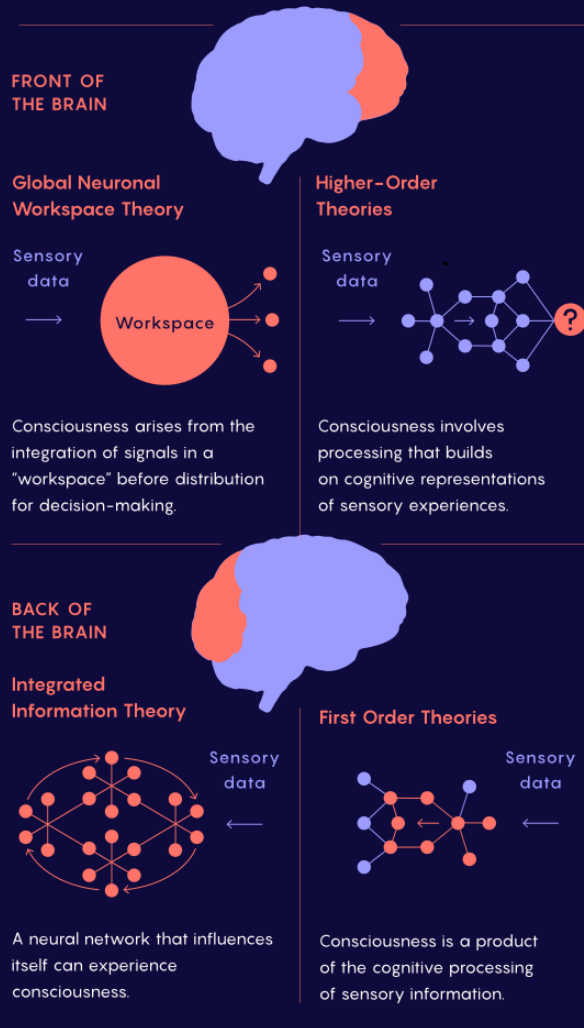
Challenges in interpreting + synthesizing highly comparative results



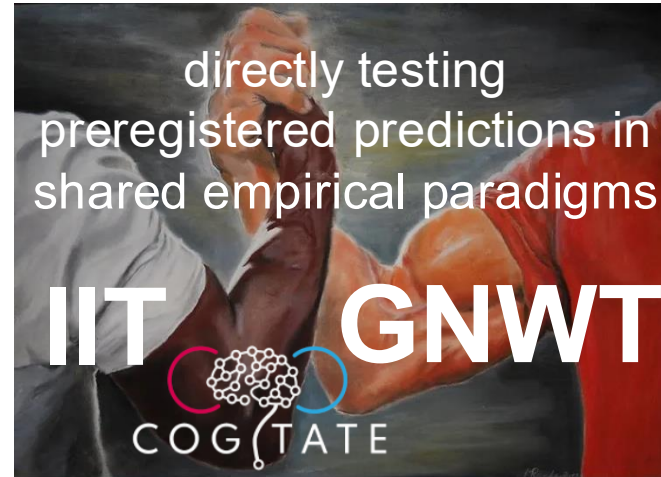
Our application: How should we measure connectivity for consciousness?

Thoughts About Consciousness

Theories about consciousness, currently numbering more than 20, can be sorted by roughly where in the brain they propose it arises. Here are some examples:



<https://www.quantamagazine.org/what-a-contest-of-consciousness-theories-really-proved-20230824/>



[nature](#) > [articles](#) > [article](#)

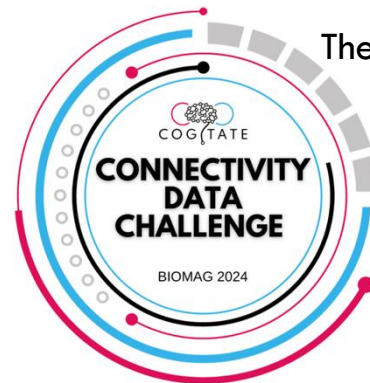
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Adversarial testing of global neuronal workspace and integrated information theories of consciousness

[Cogitate Consortium](#), [Oscar Ferrante](#), [Urszula Gorska-Klimowska](#), [Simon Henin](#), [Rony Hirschhorn](#), [Aya Khalaf](#), [Alex Lepauvre](#), [Ling Liu](#), [David Richter](#), [Yamil Vidal](#), [Niccolò Bonacchi](#), [Tanya Brown](#), [Praveen Sripad](#), [Marcelo Armendariz](#), [Katarina Bendtz](#), [Tara Ghafari](#), [Dorottya Hetenyi](#), [Jay Jeschke](#), [Csaba Kozma](#), [David R. Mazumder](#), [Stephanie Montenegro](#), [Alia Seedat](#), [Abdelrahman Sharafeldin](#), [Shuijun Yang](#), ... [Lucia Melloni](#) ✉ [+ Show authors](#)

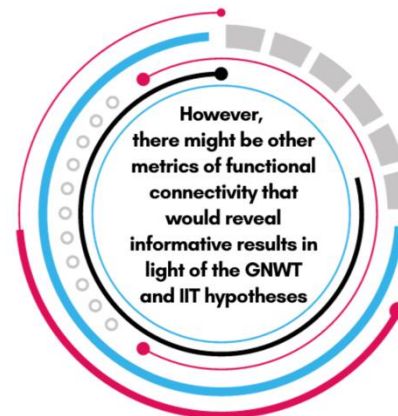
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The original FC measures, pairwise phase consistency and dynamic functional connectivity...

...did not yield evidence to support either theory

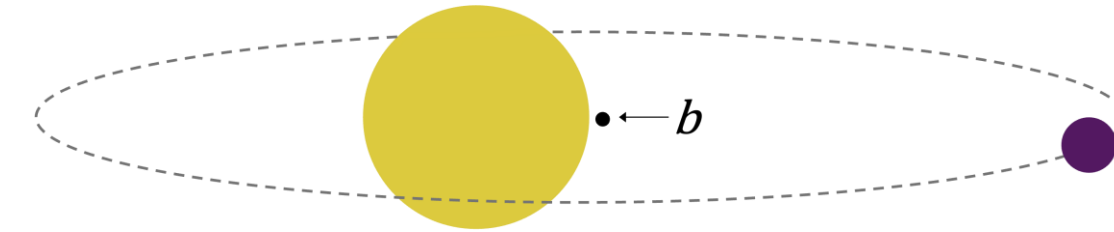
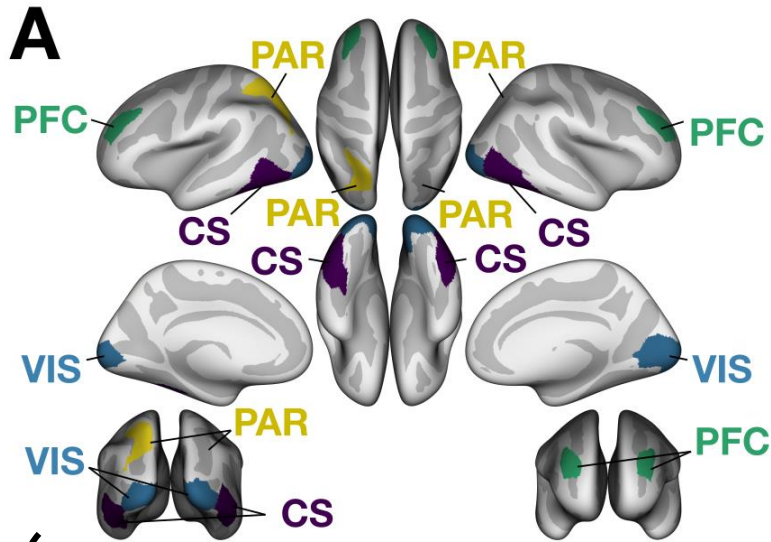


This sounds like a job for....

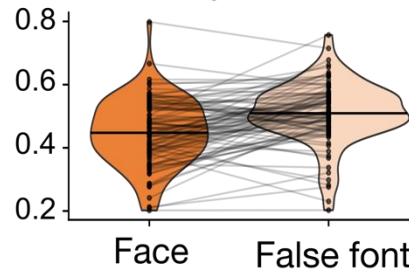
pyspi



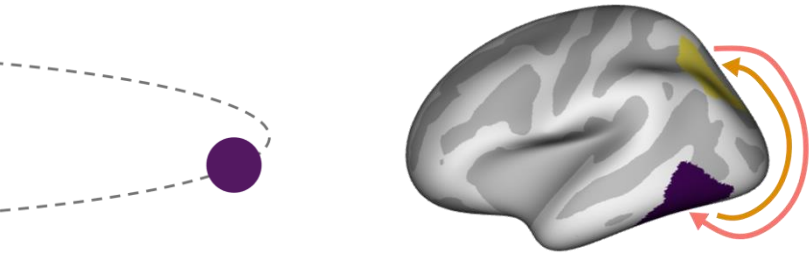
Systematic comparison highlights **barycenter**-based measures



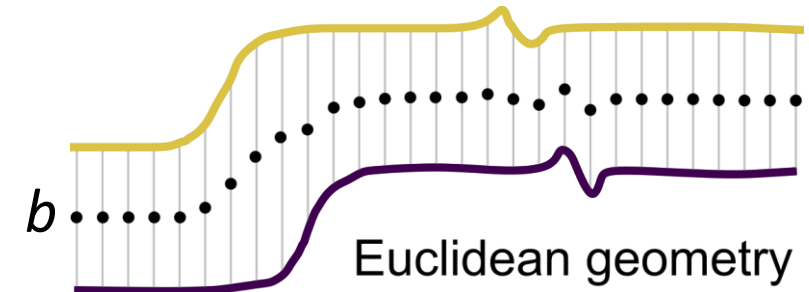
CV-accuracy: $67.0 \pm 8.3\%$



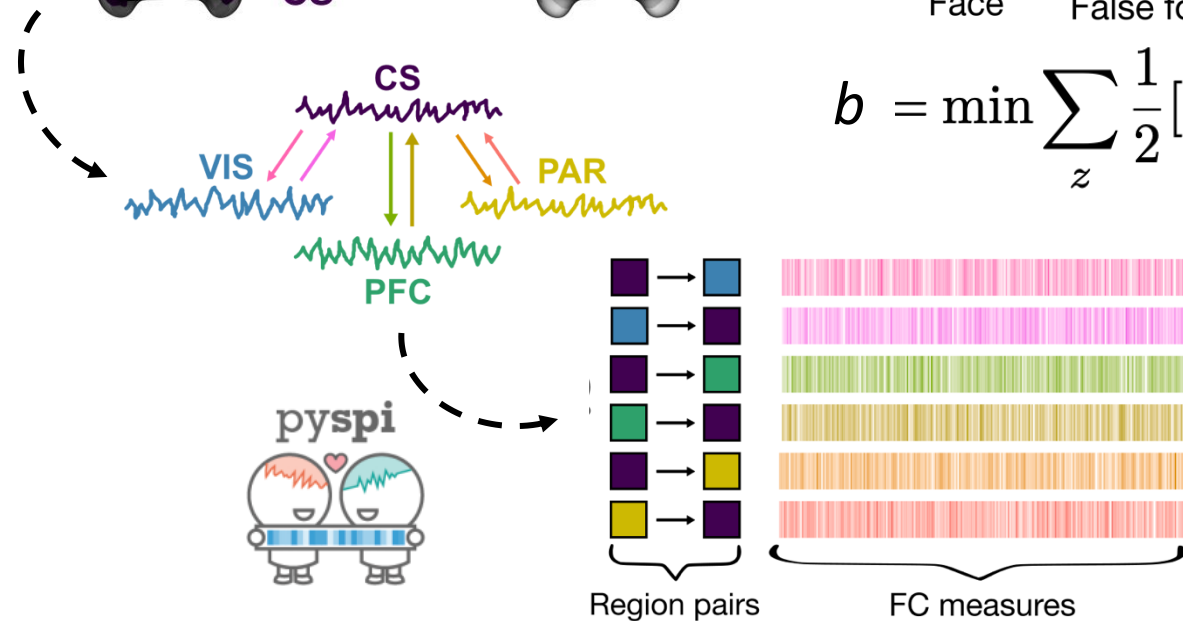
$$b = \min \sum_z \frac{1}{2} [d^2(z, x) + d^2(z, y)]$$



Category-selective (CS) —
Parietal integration area (PAR) —



💡 The **barycenter**, b , is the time series (z) that minimizes the distance, d , between our time series x and y



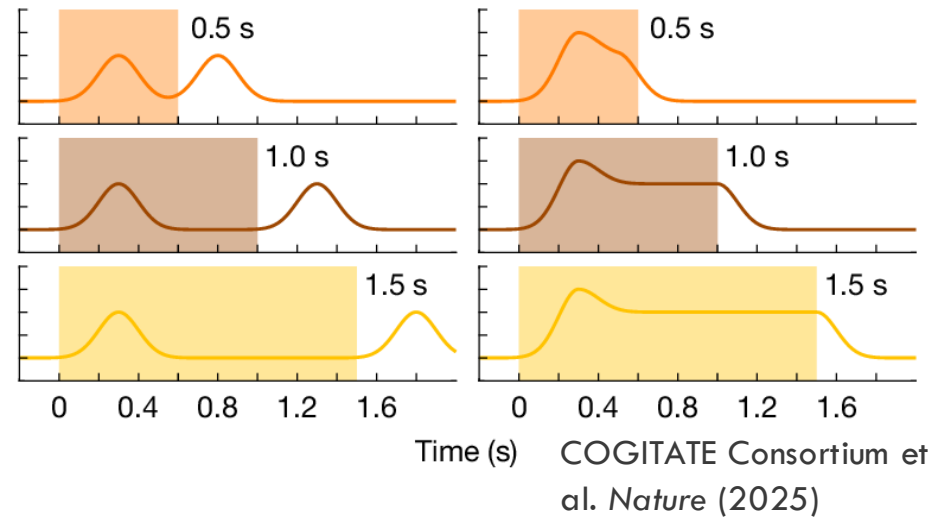
Bryant & Whyte, *Neuroscience of Consciousness* (2026)

Using **data**-driven measures to inform **hypothesis**-driven modeling

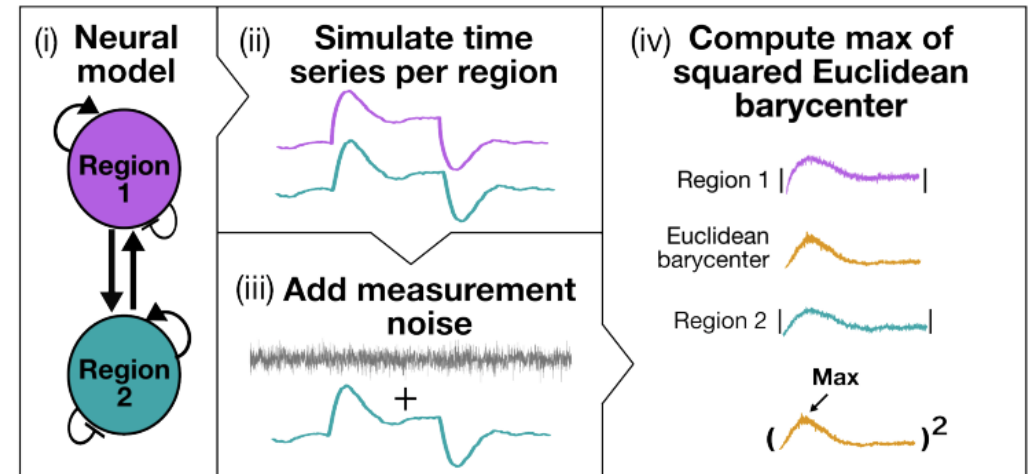
Theory-predicted neural dynamics

GNWT (Theory 1)

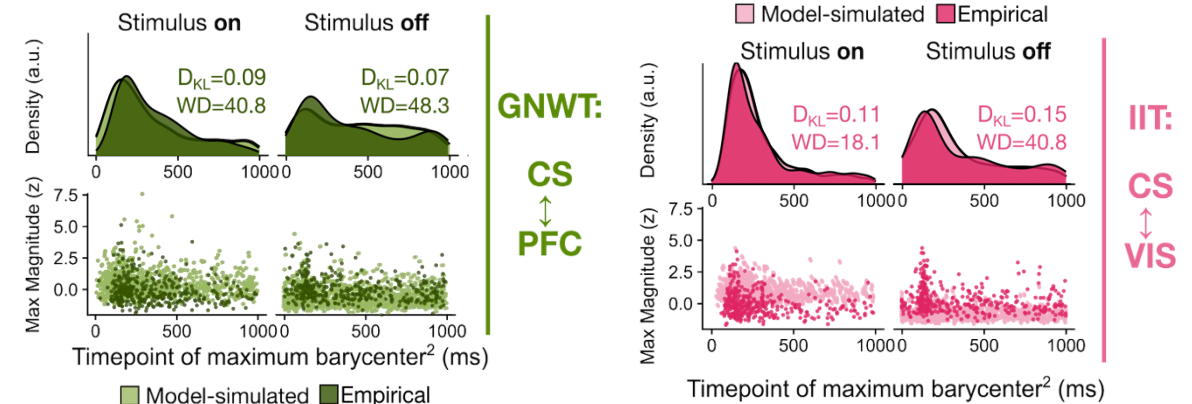
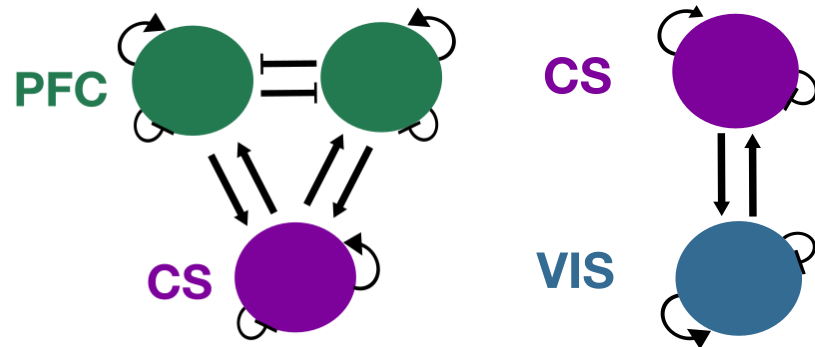
IIT (Theory 2)



Simulating theory-predicted dynamics and applying our barycenter-derived statistics to evaluate model performance

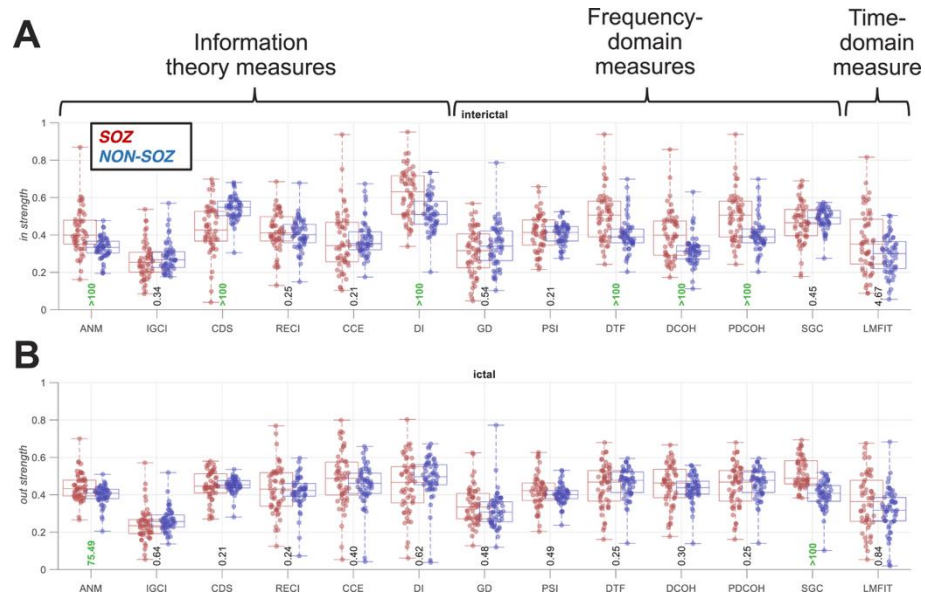
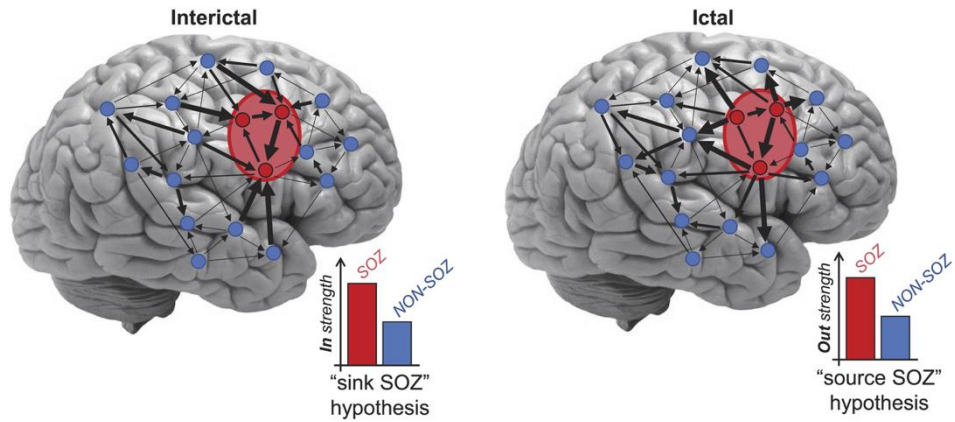


Our neural mass model implementation

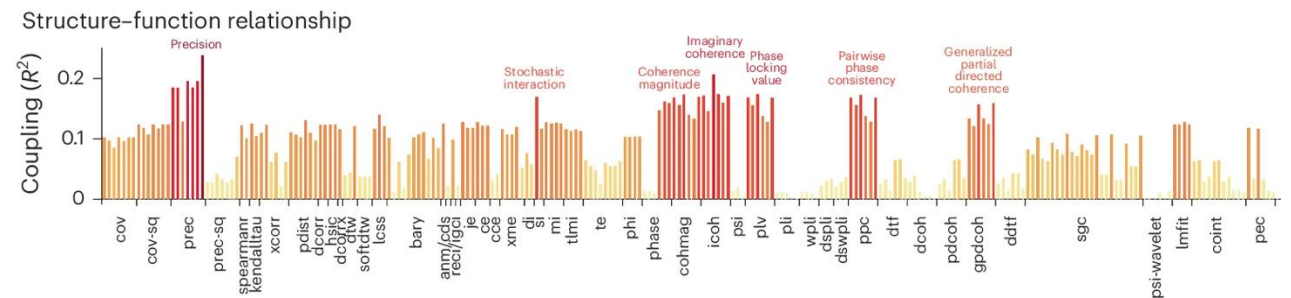
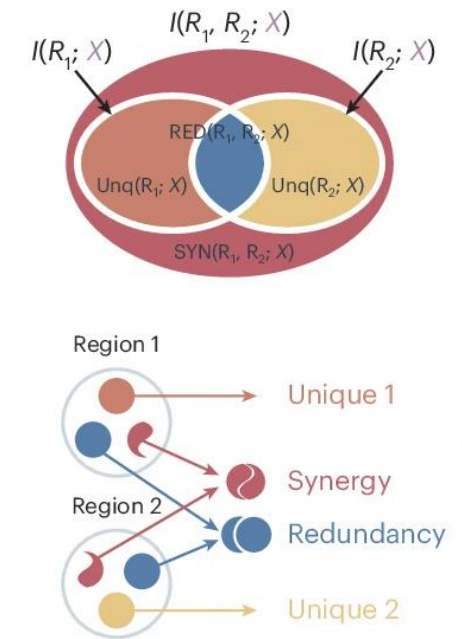
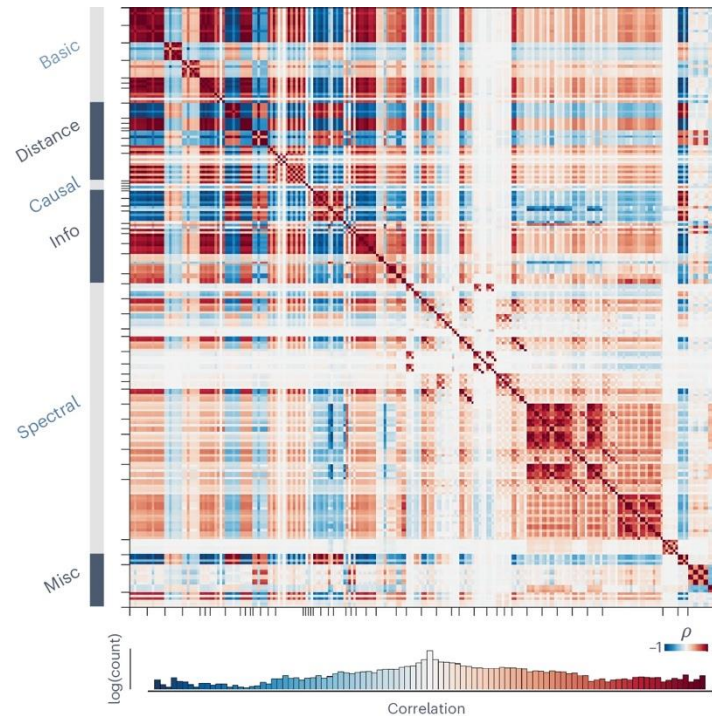


Bryant & Whyte, *Neuroscience of Consciousness* (2026)

Other groups' investigations into seizure onset + network topography



Karimi-Rouzbahani & McGonigal. *Network Neuro* (2025)



Liu et al. *Nat Methods* (2025)

Take-home messages

1

The pyspi library offers a comprehensive array of 285 statistics for pairwise interaction, enabling systematic comparison across diverse disciplines.

2

These comparisons are highly informative in computational neuroscience, highlighting unique aspects of inter-hemispheric coupling and capturing salient signal for conscious visual perception.

3

Important considerations include feature-set dimensionality, multiple comparisons correction, and static nature of scalar SPLs.

4

pyspi is actively maintained as an open-source software package, and we always welcome feedback and contributions!

Thank you for your attention & the invitation today 😊

pyspi creator



**Dr. Oliver
M. Cliff**



anniebryant/MDIG2026

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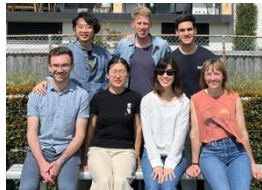
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Australian Government
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